

ARM Physical IP, Inc.

TSMC 90nm (CLN90G) Advantage Dual-Port SRAM Generator Release Notes

2005q4v1 (2005/11/21)

This is a mandatory update.

ARM Update Recommendation:

Install version 2005q4v1; discontinue using earlier versions.

Impact of this Release on Major Library Components:

- o The generated gds2 is changed in this release. The maximum power supply ring segment width was decreased for ring compliance with maximum metal density DRC. Obsolete GDS layer 121 was eliminated. Some minor layout edits were done for improved DFM.
- o The generated P&R footprint is changed in this release. Only for the case of ring power supply with ring width greater than 9um, the generated power ring structure is changed.
- o The generated timing data are unchanged in this release.
- o The generated power data are unchanged in this release.
- o All current or power values are now specified in scientific notation unless the exponent is greater than or equal to 0. This change affects the Synopsys models, ASCII datatables and PostScript datasheets.
- o The read, write and average current and power values no longer include the leakage components.
- o There is a new command line and GUI option (-asvm) that defines Verilog and VHDL model behavior in case of a setup violation on the address lines during a read operation. Options are On or Off. When not specified, the default setting is Off.
 - asvm Off
Address Setup Violation during Read: Read fails, X output;
X-out all memory locations.
 - asvm On
Address Setup Violation during Read: Read fails, X output;
All memory locations preserve their state.
- o There is a new command line and GUI option (-dpccm) that defines Verilog and VHDL model behavior in case of a clock collision. Clock collision is said to occur when the read and write addresses point to the same location AND the read and write clocks are not separated by the amount specified by Tcc, the clock collision time. Options are On or Off. When not specified, the default setting is On. The model behavior for the dpccm settings is as follows:
 - dpccm Off
Clock Collision: Read fails, X output; Write fails, X memory location and write-through (if applicable).
 - dpccm On
Clock Collision: Read fails, X output; Write to memory location succeeds.
- o The use of the same metal layer for both horizontal and vertical segments of the power supply rings has been disallowed.
- o The process of updating our product development infrastructure to change our new corporate name from "Artisan Components, Inc." to "ARM Physical IP, Inc." is in progress, and this product release reflects this change. You may still find instances of the company name "Artisan Components, Inc." or "Artisan" in these product files. This is not an error but is an artifact of the logistical difficulties

in updating all aspects of our product development infrastructure simultaneously. Note that the Artisan brand name remains a legitimate product description, and continues to be used where appropriate.

Process Technology Data

This section lists all of the foundry documents and technology files used to design this product.

- o SRAM 8T 2.24um x 0.89um (1.9936um²) Cell
 - N90DP_cell199array4x32_v1D0.gds, N90DP_cell199array4x32_v1D0_T.gds, received 2004/07/26.
 - Qualified by TSMC using deviated layout rules.
- o Design Rule Document
 - T-N90-LO-DR-001, TSMC 90nm CMOS Logic Design Rule (G/ GT/ LP), v1.4, dated 2005/06/09.
- o GDS Layer Usage Description File
 - T-N90-LO-LE-001, TSMC 90nm CAD GDS Layer Usage Description File, v1.3a, dated 2005/05/03.
- o DRC Command Files
 - CLN90S_5M.14a, TSMC 90nm CMOS Logic Process 1P5M DRC Command File, v1.4a, dated 2005/06/01.
 - CLN90S_5M.ANT.14a, CLN90G 1P5M Antenna DRC Command File, v1.4a, dated 2005/06/01.
- o LVS Command File
 - LVS_RC_Calibre_90nm_combo_deck_1p5m.13b, TSMC 90nm Calibre LVS/RC Command File, v1.3b, dated 2005/06/22.
- o Process Spice Models (used with Synopsys HSPICE -- U-2003.09-SP1)
 - Logic Models
 - Document: T-N90-LO-SP-002, dated 2004/08/27.
 - 90nm LOGIC SALICIDE (1P9M, 1.0V/2.5V) (CLN90G), v1.4, dated 2004/08/20.
 - Bit Cell Models
 - Document: T-N90-CL-CL-003, dated 2004/08/10.
 - CLN90G ULTRA HIGH DENSITY/Dual Port SRAM, v1.0, dated 2004/08/05.
- o Extractor Technology File
 - T-N90-LO-SP-002-B1, v1.4a, dated 2004/09/06.

EDA Support

This section lists the EDA tools and versions supported for this product release. The items indicated by the "*" were used during the QA testing of this release. This set of tools and versions corresponds to ARM EDA Package 4.0.

- * Cadence NC-Sim (Verilog)
 - 5.1
- * Synopsys VCS (Verilog)
 - 7.1
- * Mentor ModelSim (Verilog)
 - 6.0
- * Cadence NC-Sim (VHDL)
 - 5.1
- * Mentor ModelSim (VHDL)
 - 6.0
- * Simulation Model SDF Compatibility

- SDF 2.1
- * Synopsys Design Compiler
 - 2004.06
- * Cadence RTL Compiler with .lib
 - 4.1
- * Synopsys PrimeTime (Static Timing, Delay Calculation)
 - 2004.06
- o Synopsys Power Compiler
 - 2004.06
- o Synopsys PrimePowerPower
 - 2004.06
- o Synopsys JupiterXT
 - 2004.06
- o Cadence SoC Encounter/First Encounter (Design Planning)
 - 4.1
- * Cadence SoC Encounter/NanoRoute (Place & Route)
 - 4.1
- * Synopsys Astro (VCLEF, .lib input)
 - 2004.06
- * Mentor Graphics Calibre (GDSII, CDL)
 - 2003.4_9
- * Adobe Acrobat Reader (PDF Documentation)
 - 5.0
- * Sun Solaris Operating System (Generator Software)
 - 8
- * Redhat Linux Operating System (Generator Software)
 - 7.2

Technology Implementation

 This section provides information on items that may not be included in the foundry documentation.

Characterization corners

- Timing and power views are characterized at the following conditions:
 - fast-1 PVT = FF/1.1V/-40C
 - fast-2 PVT = FF/1.1V/0C
 - typical PVT = TT/1.0V/25C
 - slow PVT = SS/0.9V/125C
 - power PVT = FF/1.1V/125C

The executable file name for this Memory Generator is "sram_dp_adv" and it creates a default instance named "SRAM_DP_ADV".

Mandatory tapeout layers required for foundry:

- GDS2 text layer 63:63 is allocated by TSMC for IP marking.
- *** This is a mandatory tapeout layer. ***

Layers not required for mask making:

- GDS view generator includes dummy marker layers 49,119,186, used for CAD purposes.

Methodology

This section describes design flow issues that may be applicable to this release.

o Datatable

- The datatable reports the value 999.000 for undefined EMA options.

o Synopsys/Primetime/Pearl

- Please note that SDF generated with tool versions above also require postprocessing prior to back-annotation. The following postprocessing scripts are available:
 - Synopsys Design Compiler
aci/<product>/lib/sdf_postprocessors/dc_postprocessor.pl
 - Synopsys Primetime
aci/<product>/lib/sdf_postprocessors/pt_postprocessor.pl
 - Cadence Pearl
aci/<product>/lib/sdf_postprocessors/pearl_postprocessor.pl
- Please note that SDF generated with versions other than those specified above may not annotate properly due to various changes in the SDF output format.

o Technology LEF

- If ring-style power interface is specified with rings sufficiently wide to exceed 70% maximum metal density on a 100umx100um window, then the ring area should be included in chip-level metal slotting.
- The technology header information is not automatically appended by the generator. If needed, please append this information from the standard cell release.

o Antenna Modeling

- VCLEF view generator also creates the <instance_name>_ant.clf file.
- A minimum gate area as allowed by the design rules is assumed to be connected to each input pin to force the router to connect to the input pin with an antenna-clean wire. Pin connections internal to the memory have been verified to be antenna-clean.

Known Problems and Solutions

This section describes known problems and work-arounds as of the release date of this library.

- o Calibre LVS and HLVS may flag the ERC error "ERC PATHCHK GROUND && ! POWER" for some instances. This error is due to nets which have been purposely connected to ground and not power. These may be ignored. Calibre LVS and HLVS may flag the ERC error "ERC PATHCHK POWER && ! GROUND" for some instances. This error is due to nets which have been purposely connected to power and not ground. These may be ignored.

2005q1v2 (2005/03/11)

This is an optional update.

Artisan Update Recommendation:

Install version 2005q1v2; discontinue using version 2005q1v1 for new

design starts.

Impact of this Release on Major Library Components:

- o The generated gds2 is unchanged in this release.
- o The generated P&R footprint is unchanged in this release; however, the VCLEF view generator now incorporates pin antenna data in the .vclef file, rather than splitting it out into a separate antenna .lef file.
- o The generated timing data are unchanged in this release.
- o The generated power data are unchanged in this release.

Process Technology Data

This section lists all of the foundry documents and technology files used to design this product.

- o SRAM 8T 2.24um x 0.89um (1.9936um²) Cell
 - N90DP_cell199array4x32_v1D0.gds, N90DP_cell199array4x32_v1D0_T.gds, received 2004/07/26.
 - Qualified by TSMC using deviated layout rules.
- o Design Rule Document
 - TN90LODR001C1_1_3A.pdf, 90nm CMOS Logic Design Rule, v1.3A, dated 2004/10/29.
- o GDS Layer Usage Description File
 - TN90LOLE001_02A.pdf, GDS Layer Usage Description File, v0.2A, dated 2003/01/21.
- o DRC Command Files
 - CLN90S_5M.13a, Calibre DRC Command File, v1.3a, dated 2004/10/18.
 - CLN90S_6M.13a, Calibre DRC Command File, v1.3a, dated 2004/10/18.
 - CLN90S_7M.13a, Calibre DRC Command File, v1.3a, dated 2004/10/18.
 - CLN90S_8M.13a, Calibre DRC Command File, v1.3a, dated 2004/10/18.
 - CLN90S_9M.13a, Calibre DRC Command File, v1.3a, dated 2004/10/18.
 - CLN90S_5M.ANT.13a, Calibre Antenna DRC File, v1.3a, dated 2004/10/18.
 - CLN90S_6M.ANT.13a, Calibre Antenna DRC File, v1.3a, dated 2004/10/18.
 - CLN90S_7M.ANT.13a, Calibre Antenna DRC File, v1.3a, dated 2004/10/18.
 - CLN90S_8M.ANT.13a, Calibre Antenna DRC File, v1.3a, dated 2004/10/18.
 - CLN90S_9M.ANT.13a, Calibre Antenna DRC File, v1.3a, dated 2004/10/18.
- o LVS Command File
 - TSMC 90nm CALIBRE ERC/LVS COMMAND FILE, V1.1b, dated 2004/03/23.
- o Process Spice Models (used with Synopsys HSPICE -- U-2003.09-SP1)
 - Logic Models
 - Document: T-N90-LO-SP-002, dated 2004/08/27.
 - 90nm LOGIC SALICIDE (1P9M, 1.0V/2.5V) (CLN90G), V1.4, dated 2004/08/20.
 - Bit Cell Models
 - Document: T-N90-CL-CL-003, dated 2004/08/10.
 - CLN90G ULTRA HIGH DENSITY/Dual Port SRAM, V1.0, dated 2004/08/05.
- o Extractor Technology File
 - T-N90-LO-SP-002-B1, V1.4A, dated 2004/09/06.

EDA Views

This section lists the EDA tools and versions supported for this product release. The items indicated by the "*" were used during the QA testing of this release. This set of tools and versions corresponds to Artisan EDA Package 3.1-90nm.

- * Cadence NC-Sim (Verilog)
 - 3.4
- * Synopsys VCS
 - 6.2
- o Mentor ModelSim (Verilog)
 - 5.7d
- o Cadence NC-Sim (VHDL)
 - 3.4
- * Mentor ModelSim (VHDL)
 - 5.7d
- * Simulation Model SDF Compatibility
 - SDF 2.1
- * Synopsys Design Compiler (Synthesis, Delay Calculation)
 - 2002.05
- * Cadence BuildGates/PKS
 - 5.0.8
- * Synopsys PrimeTime (Static Timing, Delay Calculation)
 - 2002.03-SP1
- * Cadence NanoRoute/SOC
 - 2.01/3.1
- o Synopsys Astro (VCLEF input)
 - 2003.06
- * Mentor Graphics Calibre (GDSII, CDL)
 - 8.8(A)
- o Synopsys Power Compiler
 - 2002.05
- o Synopsys Prime Power
 - 2002.05
- o Synopsys PrimeTime SI
 - 2002.03-SP1
- * Adobe Acrobat Reader (PDF Documentation)
 - 5.0
- * Sun Solaris Operating System (Generator Software)
 - 8 (2.8)
- * Redhat Linux Operating System (Generator Software)
 - 7.2

Technology Implementation

This section provides information on items that may not be included in the foundry documentation.

- o Characterization corners
 - Timing and power views are characterized at the following conditions:
 - fast-1 PVT = FF/1.1V/-40C
 - fast-2 PVT = FF/1.1V/0C
 - typical PVT = TT/1.0V/25C
 - slow PVT = SS/0.9V/125C

power PVT = FF/1.1V/125C

- o Mandatory tapeout layers required for foundry
 - GDS2 text layer 63:63 is allocated by TSMC for IP marking.
 - *** This is a mandatory tapeout layer ***

Methodology

This section describes design flow issues that may apply to this release

- o Datatable
 - The datatable reports the value 999.000 for undefined EMA options.
- o Synopsys/Primetime
 - Please note that SDF generated with tool versions above also require post-processing prior to back-annotation. The following post-processing scripts are available:
 - Synopsys Design Compiler
aci/<product>/lib/sdf_postprocessors/dc_postprocessor.pl
 - Synopsys Primetime
aci/<product>/lib/sdf_postprocessors/pt_postprocessor.pl
 - Cadence BuildGates
aci/<product>/lib/sdf_postprocessors/bg_postprocessor.pl
 - Please note that SDF generated with versions other than those specified above may not annotate properly due to various changes in the SDF output format.
- o Technology LEF
 - The technology header information is not automatically appended by the generator. If needed, please append this information from another source.
- o Antenna Modeling
 - VCLEF view generator also creates <instance_name>_ant.clf file.
 - A minimum gate area and a minimum diffusion area as allowed by the design rules are assumed to be connected to each input pin to force the router to connect to the input pin with an antenna-clean wire. Pin connections internal to the memory have been verified to be antenna-clean. When providing antenna protection for the external wire, user must take into account the wire internal to the memory instance.

Known Problems and Solutions

This section describes known problems and work-arounds as of the release date of this library.

- o Calibre LVS and HLVS may flag the ERC error "ERC PATHCHK GROUND && ! POWER" for some instances. This error is due to nets which have been purposely connected to ground and not power. These may be ignored.
- o Calibre LVS and HLVS may flag the ERC error "ERC PATHCHK POWER && ! GROUND" for some instances. This error is due to nets which have been purposely connected to power and not ground. These may be ignored.

2005q1v1 (2005/02/11)

This is a mandatory update.

Artisan Update Recommendation:

Install version 2005q1v1; discontinue using earlier versions.

Impact of this Release on Major Library Components:

- o The generated gds2 is changed in this release. Circuit changes were made to adjust input signal set-up times.
- o The generated P&R footprint is unchanged in this release; however, the data for antenna diodes on input pins have been corrected in the generated *_ant.clf and *_ant.lef files.
- o The generated timing data are changed in this release. Circuit changes have been made and a correction was made in the characterization of Address set-up time for mux4 and mux8 cases.
- o The generated power data are changed in this release. Circuit changes have been made and a small correction was made in the mux16 current characterization.
- o The user-data validity checker was changed to check for the correct word increment.
- o The User Guide and Test & Repair Application Note were revised to clarify the Flex Repair redundancy description and correct the GUI screenshot.

Process Technology Data

This section lists all of the foundry documents and technology files used to design this product.

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- o GDS Layer Usage Description File
 - TN90LOLE001_02A.pdf, GDS Layer Usage Description File, v0.2A, dated 2003/01/21.
- o DRC Command Files
 - CLN90S_5M.13a, Calibre DRC Command File, v1.3a, dated 2004/10/18.
 - CLN90S_6M.13a, Calibre DRC Command File, v1.3a, dated 2004/10/18.
 - CLN90S_7M.13a, Calibre DRC Command File, v1.3a, dated 2004/10/18.
 - CLN90S_8M.13a, Calibre DRC Command File, v1.3a, dated 2004/10/18.
 - CLN90S_9M.13a, Calibre DRC Command File, v1.3a, dated 2004/10/18.
 - CLN90S_5M.ANT.13a, Calibre Antenna DRC File, v1.3a, dated 2004/10/18.
 - CLN90S_6M.ANT.13a, Calibre Antenna DRC File, v1.3a, dated 2004/10/18.
 - CLN90S_7M.ANT.13a, Calibre Antenna DRC File, v1.3a, dated 2004/10/18.
 - CLN90S_8M.ANT.13a, Calibre Antenna DRC File, v1.3a, dated 2004/10/18.
 - CLN90S_9M.ANT.13a, Calibre Antenna DRC File, v1.3a, dated 2004/10/18.
- o LVS Command File

- TSMC 90nm CALIBRE ERC/LVS COMMAND FILE, V1.1b, dated 2004/03/23.
- o Process Spice Models (used with Synopsys HSPICE -- U-2003.09-SP1)
 - Logic Models
 - Document: T-N90-LO-SP-002, dated 2004/08/27.
 - 90nm LOGIC SALICIDE (1P9M, 1.0V/2.5V) (CLN90G), V1.4, dated 2004/08/20.
 - Bit Cell Models
 - Document: T-N90-CL-CL-003, dated 2004/08/10.
 - CLN90G ULTRA HIGH DENSITY/Dual Port SRAM, V1.0, dated 2004/08/05.
- o Extractor Technology File
 - T-N90-LO-SP-002-B1, V1.4A, dated 2004/09/06.

EDA Views

This section lists the EDA tools and versions supported for this product release. The items indicated by the "*" were used during the QA testing of this release. This set of tools and versions corresponds to Artisan EDA Package 3.1-90nm.

- * Cadence NC-Sim (Verilog)
 - 3.4
- * Synopsys VCS
 - 6.2
- o Mentor ModelSim (Verilog)
 - 5.7d
- o Cadence NC-Sim (VHDL)
 - 3.4
- * Mentor ModelSim (VHDL)
 - 5.7d
- * Simulation Model SDF Compatibility
 - SDF 2.1
- * Synopsys Design Compiler (Synthesis, Delay Calculation)
 - 2002.05
- * Cadence BuildGates/PKS
 - 5.0.8
- * Synopsys PrimeTime (Static Timing, Delay Calculation)
 - 2002.03-SP1
- * Cadence NanoRoute/SOC
 - 2.01/3.1
- o Synopsys Astro (VCLEF input)
 - 2003.06
- * Mentor Graphics Calibre (GDSII, CDL)
 - 8.8(A)
- o Synopsys Power Compiler
 - 2002.05
- o Synopsys Prime Power
 - 2002.05
- o Synopsys PrimeTime SI
 - 2002.03-SP1
- * Adobe Acrobat Reader (PDF Documentation)
 - 5.0
- * Sun Solaris Operating System (Generator Software)
 - 8 (2.8)
- * Redhat Linux Operating System (Generator Software)
 - 7.2

Technology Implementation

This section provides information on items that may not be included in the foundry documentation.

o Characterization corners

- Timing and power views are characterized at the following conditions:
 - fast-1 PVT = FF/1.1V/-40C
 - fast-2 PVT = FF/1.1V/0C
 - typical PVT = TT/1.0V/25C
 - slow PVT = SS/0.9V/125C
 - power PVT = FF/1.1V/125C

o Mandatory tapeout layers required for foundry

- GDS2 text layer 63:63 is allocated by TSMC for IP marking.
*** This is a mandatory tapeout layer ***

Methodology

This section describes design flow issues that may apply to this release

o Datable

- The datatable reports the value 999.000 for undefined EMA options.

o Synopsys/Primetime

- Please note that SDF generated with tool versions above also require post-processing prior to back-annotation. The following post-processing scripts are available:
 - Synopsys Design Compiler
aci/<product>/lib/sdf_postprocessors/dc_postprocessor.pl
 - Synopsys Primetime
aci/<product>/lib/sdf_postprocessors/pt_postprocessor.pl
 - Cadence BuildGates
aci/<product>/lib/sdf_postprocessors/bg_postprocessor.pl
- Please note that SDF generated with versions other than those specified above may not annotate properly due to various changes in the SDF output format.

o Technology LEF

- The technology header information is not automatically appended by the generator. If needed, please append this information from another source.

o Antenna Modeling

- VCLEF view generator also creates <instance_name>_ant.lef and <instance_name>_ant.clf files.
- A minimum gate area and a minimum diffusion area as allowed by the design rules are assumed to be connected to each input pin to force the router to connect to the input pin with an antenna-clean wire. Pin connections internal to the memory have been verified to be antenna-clean. When providing antenna protection for the external wire, user must take into account the wire internal to the memory instance.

Known Problems and Solutions

This section describes known problems and work-arounds as of the release date of this library.

- o Calibre LVS and HLVS may flag the ERC error "ERC PATHCHK GROUND && ! POWER" for some instances. This error is due to nets which have been purposely connected to ground and not power. These may be ignored.
- o Calibre LVS and HLVS may flag the ERC error "ERC PATHCHK POWER && ! GROUND" for some instances. This error is due to nets which have been purposely connected to power and not ground. These may be ignored.

2004q4v2 (2004/12/30)

This is a mandatory update.

Artisan Update Recommendation:

Install version 2004q4v2; re-evaluate timing and power.
Discontinue using version 2004q4v1.

Impact of this Release on Major Library Components:

- o The generated gds2 is changed in this release. Circuit changes were made for clean margin analysis using updated extraction flow and Spice models; circuit changes were made to adjust for EMA default correction; minor layout changes were made for compliance with updated DRC deck.
- o The generated P&R footprint is unchanged in this release.
- o The generated timing data are changed in this release. Circuit changes have been made and the generator has been recharacterized based upon updated extraction flow and Spice models.
- o The generated power data are changed in this release. Circuit changes have been made and the generator has been recharacterized based upon updated extraction flow and Spice models.

Process Technology Data

This section lists all of the foundry documents and technology files used to design this product.

- o SRAM 8T 2.24um x 0.89um (1.9936um²) Cell
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- CLN90S_5M.ANT.13a, Calibre Antenna DRC File, v1.3a, dated 2004/10/18.
- CLN90S_6M.ANT.13a, Calibre Antenna DRC File, v1.3a, dated 2004/10/18.
- CLN90S_7M.ANT.13a, Calibre Antenna DRC File, v1.3a, dated 2004/10/18.
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- CLN90S_9M.ANT.13a, Calibre Antenna DRC File, v1.3a, dated 2004/10/18.
- o LVS Command File
 - TSMC 90nm CALIBRE ERC/LVS COMMAND FILE, V1.1b, dated 2004/03/23.
- o Process Spice Models (used with Synopsys HSPICE -- U-2003.09-SP1)
 - Logic Models
 - Document: T-N90-LO-SP-002, dated 2004/08/27.
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- o Extractor Technology File
 - T-N90-LO-SP-002-B1, V1.4A, dated 2004/09/06.

EDA Views

This section lists the EDA tools and versions supported for this product release. The items indicated by the "*" were used during the QA testing of this release. This set of tools and versions corresponds to Artisan EDA Package 3.1-90nm.

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- * Simulation Model SDF Compatibility
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 - 5.0.8
- * Synopsys PrimeTime (Static Timing, Delay Calculation)
 - 2002.03-SP1
- * Cadence NanoRoute/SOC
 - 2.01/3.1
- o Synopsys Astro (VCLEF input)
 - 2003.06
- * Mentor Graphics Calibre (GDSII, CDL)
 - 8.8(A)

- o Synopsys Power Compiler
 - 2002.05
- o Synopsys Prime Power
 - 2002.05
- o Synopsys PrimeTime SI
 - 2002.03-SP1
- * Adobe Acrobat Reader (PDF Documentation)
 - 5.0
- * Sun Solaris Operating System (Generator Software)
 - 8 (2.8)
- * Redhat Linux Operating System (Generator Software)
 - 7.2

Technology Implementation

This section provides information on items that may not be included in the foundry documentation.

- o Characterization corners
 - Timing and power views are characterized at the following conditions:
 - fast-1 PVT = FF/1.1V/-40C
 - fast-2 PVT = FF/1.1V/0C
 - typical PVT = TT/1.0V/25C
 - slow PVT = SS/0.9V/125C
 - power PVT = FF/1.1V/125C
- o Mandatory tapeout layers required for foundry
 - GDS2 text layer 63:63 is allocated by TSMC for IP marking.
 - *** This is a mandatory tapeout layer ***

Methodology

This section describes design flow issues that may apply to this release

- o Datatable
 - The datatable reports the value 999.000 for undefined EMA options.
- o Synopsys/Primetime
 - Please note that SDF generated with tool versions above also require post-processing prior to back-annotation. The following post-processing scripts are available:
 - Synopsys Design Compiler
aci/<product>/lib/sdf_postprocessors/dc_postprocessor.pl
 - Synopsys Primetime
aci/<product>/lib/sdf_postprocessors/pt_postprocessor.pl
 - Cadence BuildGates
aci/<product>/lib/sdf_postprocessors/bg_postprocessor.pl
 - Please note that SDF generated with versions other than those specified above may not annotate properly due to various changes in the SDF output format.
- o Technology LEF
 - The technology header information is not automatically appended by the generator. If needed, please append this information

from another source.

o Antenna Modeling

- VCLEF view generator also creates <instance_name>_ant.lef and <instance_name>_ant.clf files.
- A minimum gate area and a minimum diffusion area as allowed by the design rules are assumed to be connected to each input pin to force the router to connect to the input pin with an antenna-clean wire. Pin connections internal to the memory have been verified to be antenna-clean. When providing antenna protection for the external wire, user must take into account the wire internal to the memory instance.

Known Problems and Solutions

This section describes known problems and work-arounds as of the release date of this library.

- o Calibre LVS and HLVS may flag the ERC error "ERC PATHCHK GROUND && ! POWER" for some instances. This error is due to nets which have been purposely connected to ground and not power. These may be ignored.
- o Calibre LVS and HLVS may flag the ERC error "ERC PATHCHK POWER && ! GROUND" for some instances. This error is due to nets which have been purposely connected to power and not ground. These may be ignored.

2004q4v1 (2004/11/22)

*** This is an initial release. ***

Process Technology Data

This section lists all of the foundry documents and technology files used to design this product.

- o SRAM 8T 2.24um x 0.89um (1.9936um²) Cell
 - N90DP_cell199array4x32_v1D0.gds, N90DP_cell199array4x32_v1D0_T.gds, received 2004/07/26.
 - Qualified by TSMC using deviated layout rules.
- o Design Rule Document
 - TN90LODR001_1_2.pdf, CMOS Logic Design Rule, v1.2, dated 2004/04/06.
- o GDS Layer Usage Description File
 - TN90LOLE001_02A.pdf, GDS Layer Usage Description File, v0.2A, dated 2003/01/21.
- o DRC Command Files
 - CLN90S_5M.12a.1, Calibre DRC Command File, v1.2a.1, dated 2004/03/15.
 - CLN90S_6M.12a.1, Calibre DRC Command File, v1.2a.1, dated 2004/03/15.
 - CLN90S_7M.12a.1, Calibre DRC Command File, v1.2a.1, dated 2004/03/15.
 - CLN90S_8M.12a.1, Calibre DRC Command File, v1.2a.1, dated 2004/03/15.
 - CLN90S_9M.12a.1, Calibre DRC Command File, v1.2a.1, dated 2004/03/15.
 - CLN90S_5M.ANT.12a.1, Calibre Antenna DRC File, v1.2a.1, 2004/03/15.
 - CLN90S_6M.ANT.12a.1, Calibre Antenna DRC File, v1.2a.1, 2004/03/15.

CLN90S_7M.ANT.12a.1, Calibre Antenna DRC File, v1.2a.1, 2004/03/15.
CLN90S_8M.ANT.12a.1, Calibre Antenna DRC File, v1.2a.1, 2004/03/15.
CLN90S_9M.ANT.12a.1, Calibre Antenna DRC File, v1.2a.1, 2004/03/15.

o LVS Command File

- TSMC 90nm CALIBRE ERC/LVS COMMAND FILE, V1.1b, dated 2004/03/23.

o Process Spice Models (used with Synopsys HSPICE -- U-2003.09-SP1)

- 90nm LOGIC SALICIDE (1P9M, 1.0V/2.5V) (CLN90G), V1.2, dated 2004/02/13.
- CLN90G DUAL PORT SRAM, V0.02, dated 2004/01/02.

o Extractor Technology File

- T-N90-L0-SP-002-B1, V1.2A, dated 2004/03/04.

EDA Views

This section lists the EDA tools and versions supported for this product release. The items indicated by the "*" were used during the QA testing of this release. This set of tools and versions corresponds to Artisan EDA Package 3.1-90nm.

- * Cadence NC-Sim (Verilog)
 - 3.4
- * Synopsys VCS
 - 6.2
- o Mentor ModelSim (Verilog)
 - 5.7d
- o Cadence NC-Sim (VHDL)
 - 3.4
- * Mentor ModelSim (VHDL)
 - 5.7d
- * Simulation Model SDF Compatibility
 - SDF 2.1
- * Synopsys Design Compiler (Synthesis, Delay Calculation)
 - 2002.05
- * Cadence BuildGates/PKS
 - 5.0.8
- * Synopsys PrimeTime (Static Timing, Delay Calculation)
 - 2002.03-SP1
- * Cadence NanoRoute/SOC
 - 2.01/3.1
- o Synopsys Astro (VCLEF input)
 - 2003.06
- * Mentor Graphics Calibre (GDSII, CDL)
 - 8.8(A)
- o Synopsys Power Compiler
 - 2002.05
- o Synopsys Prime Power
 - 2002.05
- o Synopsys PrimeTime SI
 - 2002.03-SP1
- * Adobe Acrobat Reader (PDF Documentation)
 - 5.0
- * Sun Solaris Operating System (Generator Software)
 - 8 (2.8)
- * Redhat Linux Operating System (Generator Software)

Technology Implementation

This section provides information on items that may not be included in the foundry documentation.

o Characterization corners

- Timing and power views are characterized at the following conditions:
 - fast-1 PVT = FF/1.1V/-40C
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 - slow PVT = SS/0.9V/125C
 - power PVT = FF/1.1V/125C

o Mandatory tapeout layers required for foundry

- GDS2 text layer 63:63 is allocated by TSMC for IP marking.
*** This is a mandatory tapeout layer ***

Methodology

This section describes design flow issues that may apply to this release

o Datable

- The datatable reports the value 999.000 for undefined EMA options.

o Synopsys/Primetime

- Please note that SDF generated with tool versions above also require post-processing prior to back-annotation. The following post-processing scripts are available:
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aci/<product>/lib/sdf_postprocessors/bg_postprocessor.pl
- Please note that SDF generated with versions other than those specified above may not annotate properly due to various changes in the SDF output format.

o Technology LEF

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o Antenna Modeling

- VCLEF view generator also creates <instance_name>_ant.lef and <instance_name>_ant.clf files.
- A minimum gate area and a minimum diffusion area as allowed by the design rules are assumed to be connected to each input pin to force the router to connect to the input pin with an antenna-clean wire. Pin connections internal to the memory have been verified to be antenna-clean. When providing antenna protection for the external wire, user must take into account

the wire internal to the memory instance.

Known Problems and Solutions

This section describes known problems and work-arounds as of the release date of this library.

- o Calibre LVS and HLVs may flag the ERC error "ERC PATHCHK GROUND && ! POWER" for some instances. This error is due to nets which have been purposely connected to ground and not power. These may be ignored.
 - o Calibre LVS and HLVs may flag the ERC error "ERC PATHCHK POWER && ! GROUND" for some instances. This error is due to nets which have been purposely connected to power and not ground. These may be ignored.
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